

MEDIATEK

MT8676 Hypervisor Camera User Manual

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Version History

Version	Date	Author	Description
1.0	2024-10-31	Jian.Xiang	Official release
1.1	2024-11-22	Jian.Xiang	Modified the description in Section 1.3.1 Architecture

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1 Hypervisor Camera

1.1 Overview

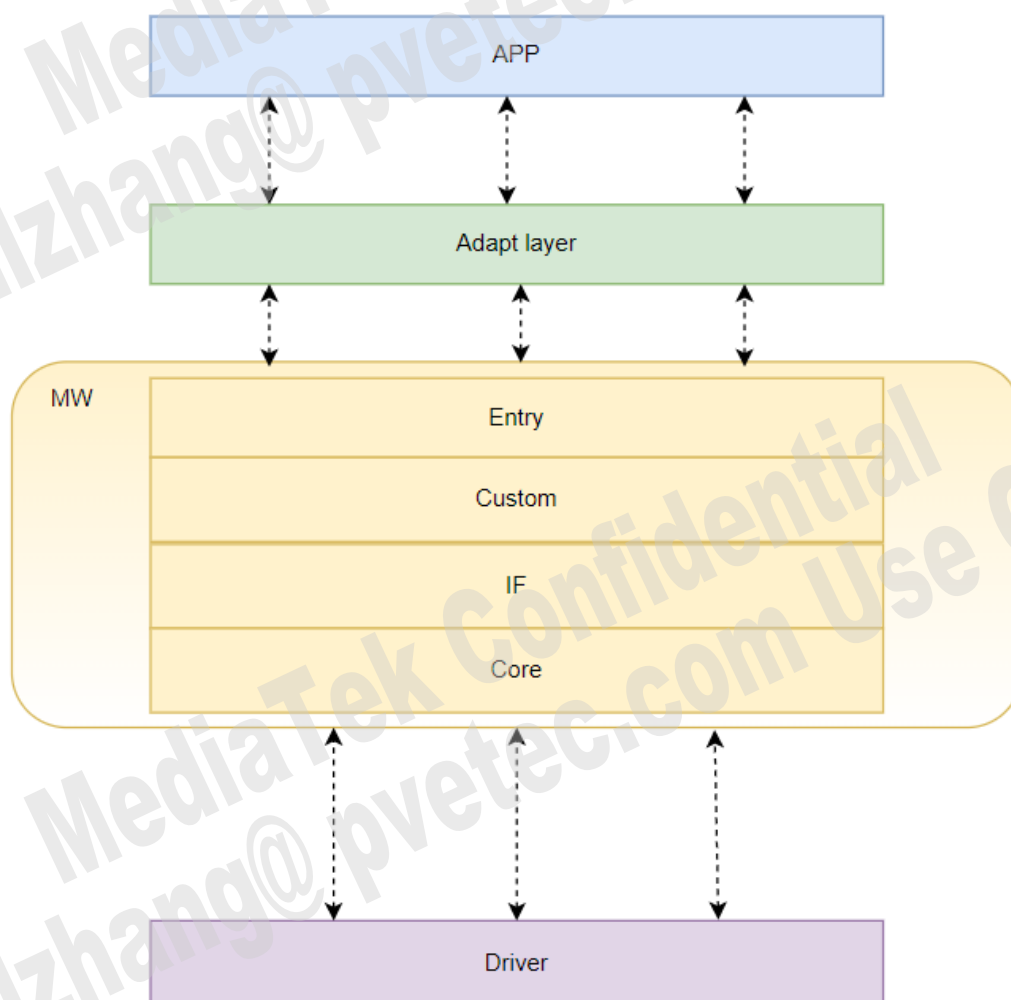


Figure 1-1. Camera architecture overview

Camera Turbo is mainly divided into three parts:

- The Entry layer is the entrance to MW. Different OSs can adapt MW by calling the code of the entry layer through the adapt layer.
- The Custom layer is the customization layer through which customers can perform customized operations.
- The IF layer is the interface layer used by the entry layer. The logic of camera session and native camera is generated here. The IF layer mainly implements the flow framework of request and config-related operations.
- The Core layer is the core layer of MW. It mainly communicates with the driver, sends the request and gets the result back, and finally calls back to the APP through the IF layer.

1.2 Camera Turbo

1.2.1 Architecture

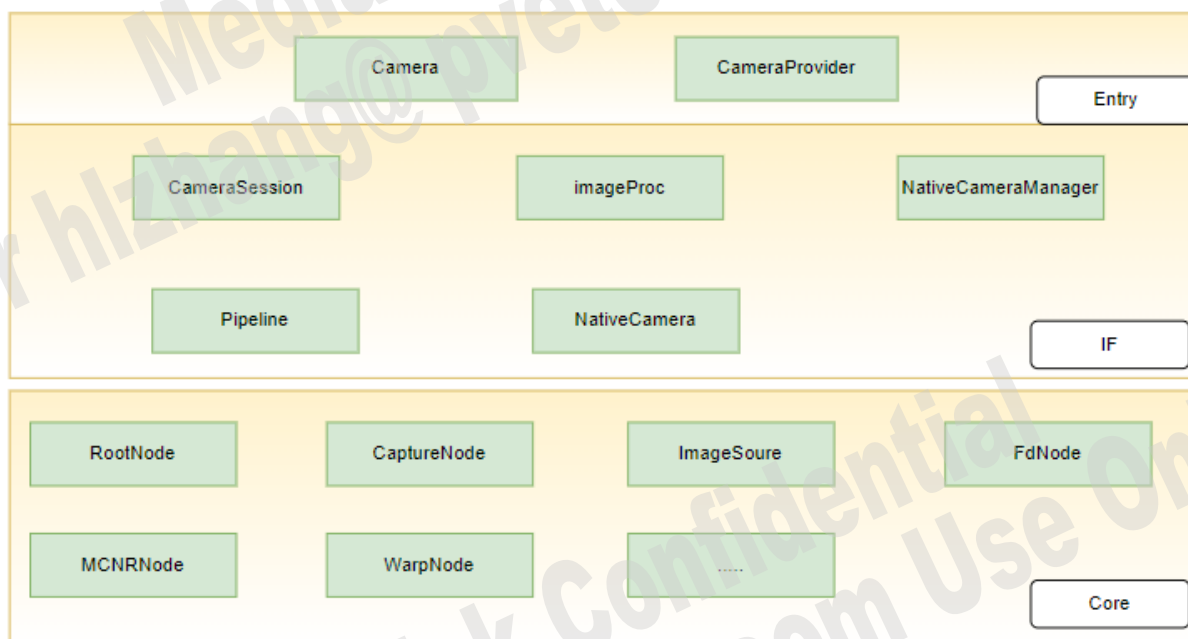


Figure 1-2. Camera MW architecture

The main architecture diagram of Camera Turbo is the Entry layer, IF layer and Core layer. The Custom layer is mainly for customer use. If not used, it can be bypassed directly.

- The Entry layer is mainly composed of Camera and CameraProvider. The CameraProvider mainly provides the Camera list to the upper layer and obtains and controls the properties of the Camera. The Camera mainly provides the interface to the upper layer to control the control behavior of the camera sensor.
- The IF layer is mainly the pipeline, which is the manager that builds the underlying node connection relationship. ImageProc mainly facilitates callback to provide customers with customized behaviors. At the same time, customers can also customize the pipeline through ImagePro.
- The Core layer is mainly composed of subclasses of ImageNode. The subclasses interact with the underlying driver to implement different operations on the request. For example, FdNode is mainly for face recognition, captureNode for Camera function, MCNRNode for image processing, and the final result will be passed through ImageNode callback to IF layer.

1.2.2 Control Flow

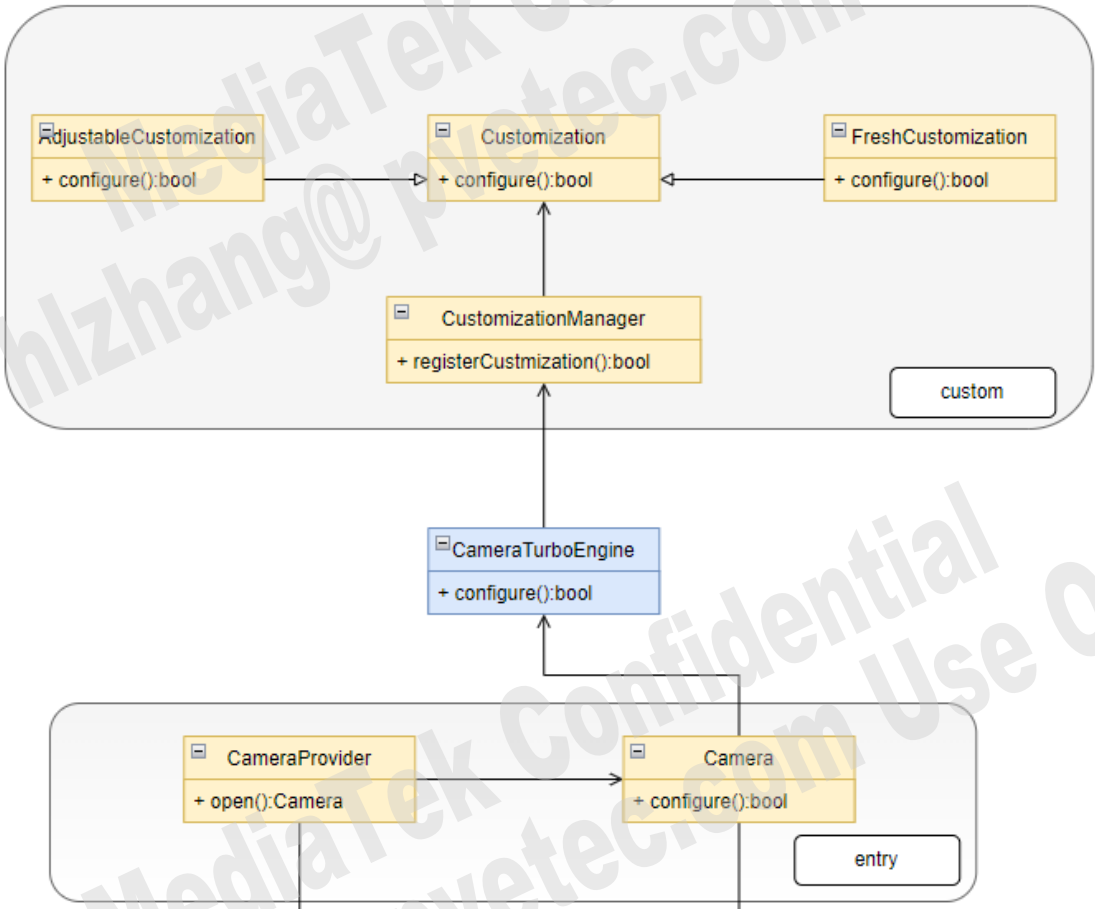


Figure 1-3. MW class flow

Figure 1-3 is mainly the class flow interaction diagram between Entry and Custom. It is mainly that the upper layer gets the camera through the open interface of CameraProvider, and the camera gets the CustomizationManger by calling Camera TurboEngine. CustomizationManger is the manager of the Custom layer, which facilitates the management of customer-specific customization behaviors.

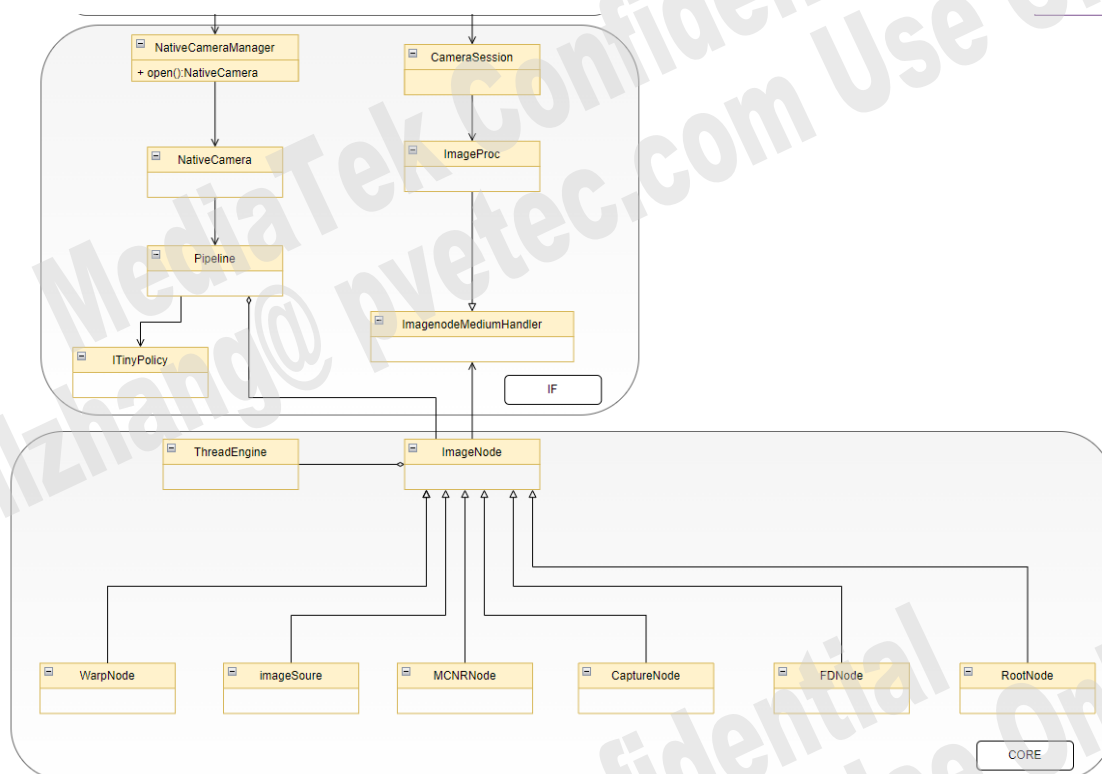


Figure 1-4. MW class flow

The interaction logic between the IF layer and Core is mainly that the open interface of the upper CameraProvider calls the NativeCameraManager. The NativeCameraManager allocates the NativeCamera that can be operated at the bottom layer. Each NativeCamera has pipeline attributes, which determines the actual flows of each NativeCamera. The camera configuration will create a new CameraSession. CameraSession creates its own Pipeline. The connected ImageProc is saved in the Pipeline. ImageProc will create the corresponding ImageNode. Each CameraSession has its own ImageProc list. Users can perform customized behaviors in ImageProc. ImageProc will use MediumHandler, which is the callback interface. ImageNode callback will use ImageProc.

1.3 Camera Virtualization

1.3.1 Architecture

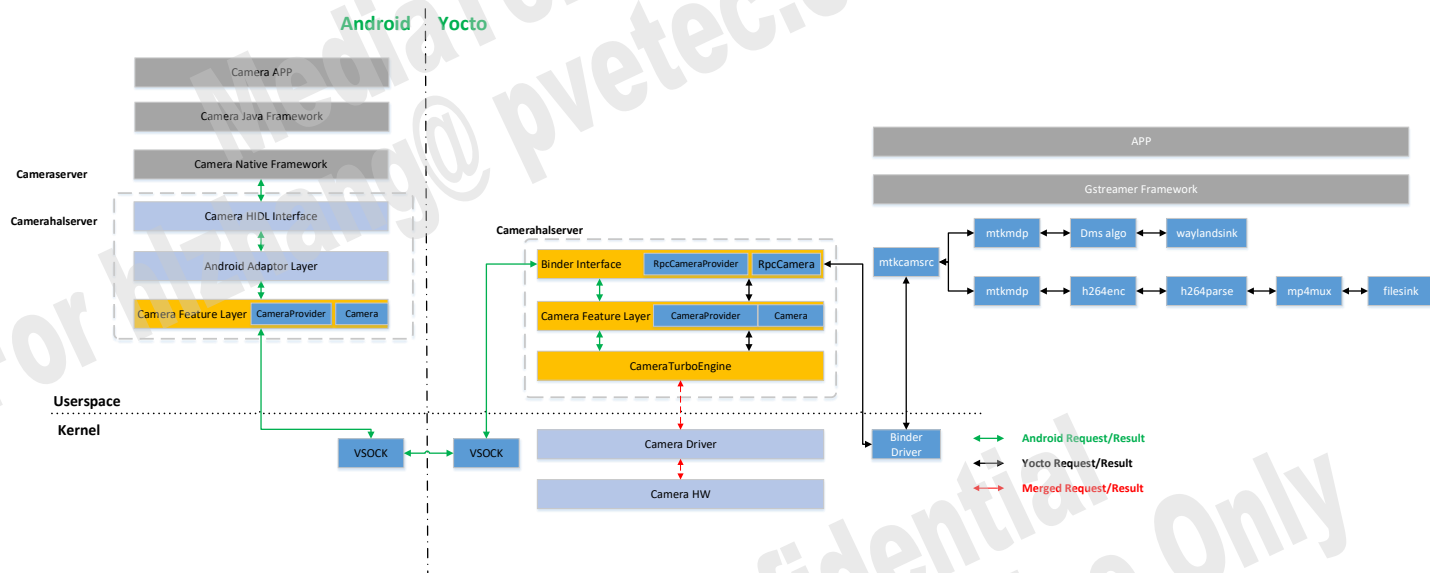


Figure 1-5. Camera Virtualization architecture

The guest (Android) and host (Yocto) systems are running simultaneously, and operations on both systems are performed separately. There is a guest camerahalservice process on the Android side, and a host camerahalservice process on the Yocto side. The APP is on the Android side, and the camera underlying is on the Yocto side (MW/sensor/driver). The camerahalservice on the Android side takes pictures from the camerahalservice on the Yocto side as a guest role. The camerahalservice on the Yocto side needs to run first so that the Android camera APP can be opened. When it is necessary to restart the camerahalservice, the Yocto side is restarted first, and then the Android side's camerahalservice is killed.

1.4 Debug Share

1.4.1 Log

```
camerahalservice: [3457][TINY_NATIVECAMERAMANAGER] open cam::0 +
camerahalservice: [3457][RpcCamera] [RpcCamera] Cam::0, userId 0, logEn 0, ctor
camerahalservice: [3995][TINY_NATIVECAMERA] [operator()] Power On Sensor Success
camerahalservice: [3457][RpcCamera] [configure] Cam::0, userId 0 +
camerahalservice: [3457][TINY_CAMERA] [configure] cam[0] user[0] configure +
camerahalservice: [3457][TINY_IMAGEPROC] [addMultiCallback] [captureProc] add
callback:0x7f6002b350 for user:0
camerahalservice: [4061][TINY_NATIVECAMERA] [0][operator()] [user::0]current pipeline
active ...
camerahalservice: [4061][TINY_PIPELINE] ppl::0, active, root 0x7f60021fd0 +
camerahalservice: [3457][TinyMW/TinyPolicy] [TinyMW/TinyPolicy::initialize] +
camerahalservice: [3457][TinyMW/TinyPolicy] [TinyMW/TinyPolicy::initialize] -
camerahalservice: [3457][TINY_PIPELINE] ppl::0, init +
camerahalservice: [3457][TINY_PIPELINE] init -
camerahalservice: [3457][TINY_NATIVECAMERA] [0][configStreams] [user::0]curpipe init -
camerahalservice: [3457][TINY_NATIVECAMERA] [0][pushRequest] [user:0 R:0] +
camerahalservice: [3457][TINY_NATIVECAMERA] [0][pushRequest] [user:0 R:0] -
camerahalservice: [3994][TinyMW/TinyPolicy] [TinyMW/TinyPolicy::onEvaluateRequest]
AppRequest: 0 in Device +
camerahalservice: [3994][TINY_ROOTNODE] [0]( 86)[enqReqLoop] user:0 frameId:0
camerahalservice: [3994][TINY_IMAGENODE] [P1Node_0]( 101)[input][Cam::0] [NODEDBG_FRAME
R:0,F:0] input
camerahalservice: [3994][TINY_IMAGENODE] [ImgSrc_256]( 101)[input][Cam::0] [NODEDBG_FRAME
R:0,F:0] input
camerahalservice: [4101][TINY_IMAGEPROC] [notify] [0][P1Proc][0x7f600395b0]notify shutter
user:0 shutter:195729495000
camerahalservice: [4101][TINY_IMAGENODE] [P1Node_0]( 370)[applyRelease][Cam::0] R0:F0
camerahalservice: [3988][TINY_IMAGENODE] [McnrNode_1]( 101)[input][Cam::0] [NODEDBG_FRAME
R:0,F:0] input
camerahalservice: [4020][TINY_IMAGENODE] [McnrNode_1]( 370)[applyRelease][Cam::0] R0:F0
camerahalservice: [4020][TINY_IMAGENODE] [McnrNode_1]( 241)[applyResult][Cam::0]
[NODEDBG_FRAME R:0,F:0] out, result(0), #processing(0)
camerahalservice: [4020][CustLayer] [processCaptureResult] processCaptureResult camId:0
userId 0, frameId:0
```

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